

ASSIGNMENT-1

Dataset Selection and Preparation

Group-208

Group Member	Main Responsibility and contribution	Agreed Contribution (%)
560441347-spada0736	Dataset audition and selection, Data readiness, Cleaning Audit	50
560299645-arya0928	Report Review, EDA, Stakeholder Problem and success criteria	50

1. Datasets Comparison Table

Dataset	Size and variables	Opportunity and stakeholder decision	Quality risk and ethical or feasibility concern
NSW Train Occupancy	50,250 rows; 14 columns; 2 numeric and 12 categorical/text	Identify crowding patterns and later predict crowding risk. NSW train operations can prioritise capacity or timetable review by time, station, and corridor.	Risk: mixed timestamp formats, minor missingness, and 0.5% duplicates. Concern: the short January observation period may not represent all demand conditions.
Air Travel Delay	101,000 rows; 29 columns; 14 numeric and 15 categorical/text	Identify delay patterns and later predict delay risk. Airline or airport operations can prioritise disruption-prevention and delay-management actions.	Risk: several delay-related fields have substantial missingness. Concern: historical operations may not generalise to future disruptions or weather events.
Hotel Booking	119,987 rows; 32 columns; 20 numeric and 12 categorical/text	Analyse cancellations and booking demand. Hotel managers can adjust staffing, bookings, and revenue-management decisions.	Risk: 26.1% exact duplicates. Concern: booking records may reflect historical customer behaviour and require care around customer-data privacy.

The NSW Train Occupancy Dataset was chosen due to its feasibility and relevance for Assignment 2. It includes 50,250 observations and 14 features, low number of missing values, and 0.5% of exact duplicates. On the other hand, the Airline Delay Dataset contains missing values in its variables that define causes of delays. Moreover, the Hotel Booking Dataset contains 26.1% of duplicates. The Train Occupancy Dataset addresses the needs of the assignment in advising an NSW transport operations team about the situation with crowds.

2. Stakeholder Problem and success criteria

The identified primary stakeholders will be the NSW trains operation team. They are responsible for ensuring timetables and services are catered to provide the most efficient and robust schedule while minimizing crowding and delays.

The team will need to decide which times, station and routes should receive priority for capacity or timetable review

Research Question: Which departure times, station stops and service line-direction/route combinations show the highest crowding risks?

Success criteria: A successful and useful answer would be able to identify the number of high priority times and locations using concrete evidence through exploratory analysis, rather than having to treat every service with equal priority.

Cost of Misleading Conclusion: A misleading conclusion will affect the real decisions made by the team. Data driven decision need to be made with evidence that is proper and correct. A false finding, in this case providing a station, time or route that doesn't need priority will lead to waste of resources that instead of solving a problem will lead to more loss. If we identify the wrong crowded routes passengers will still keep dealing with packed trains, fewer seats and more time loss. This makes travel less comfortable and reliable and can cause lack of trust or public shame to the NSW trains department. Hence, we must provide correct information and use this data as a guide to an investigation rather than final proof.

3. Cleaning Evidence

The following cleaning log provides the steps taken to reach the final dataset.

Issue	Evidence	Action	Justification	Impact
Missing occupancy status	73 records	Dropped rows with missing Occupancy Status	The target could not be reliably inferred from Occupancy Range	50,250 to 50,177 rows before duplicate removal
Exact duplicates	250 duplicate rows	Removed exact duplicates	Repeated records add no information and would distort frequencies	50,177 to 49,927 rows

Whitespace inconsistencies	Up , Down, Hunter Valley	Applied .str.strip()	Standardises equivalent labels	Categories combined
Inconsistent occupancy labels	165 MANY_SEATSAVAILABLE, 8 lowercase few-seats labels, 2 typo standing-room labels	Replaced with standard labels	Prevents equivalent categories from being counted separately	Three valid occupancy categories
Mixed timestamp formats	Full datetimes, 1421, 09.08PM, unknown, ??	Parsed valid formats; paired valid timestamp supplies date for time-only values	Avoids interpreting compact times as years	202 time-only values repaired
Unusable timestamps	19 arrival and 28 departure values remain unavailable	Retained in master file; excluded only from timestamp-complete subset	These records remain useful for non-timestamp EDA	49,880 timestamp-complete records

- The initial raw data set ‘train_occupancy.csv’ comprised of 50,250 records with 14 variables (evidence provided in ‘data_readiness. ipynb’).
- We found 250 duplicate rows that match exactly to each other and hence provide no actionable data, these were dropped from the dataset.
- On further analysis we found 73 records with missing ‘occupancy status’ and removed since they can’t be inferred from just occupancy range.
- IMPORTANT TARGET LEAKAGE: occupancy range was excluded since it is closely related to occupancy status and occupancy status provides a much clearer image for predictive analysis
- String cleaning was performed to remove whitespace from categorical variable (categories: UP, Down and Hunter Valley) also combining the categories.
- Next, we standardize the occupancy labels: MANY_SEATS_AVAILABLE, FEW_SEATS_AVAILABLE and STANDING_ROOM_ONLY.
- Numerical fields were checked and found that ‘day’ ranges from 9-16 and ‘Node.seq.order’ ranges from 1-38 without invalid values

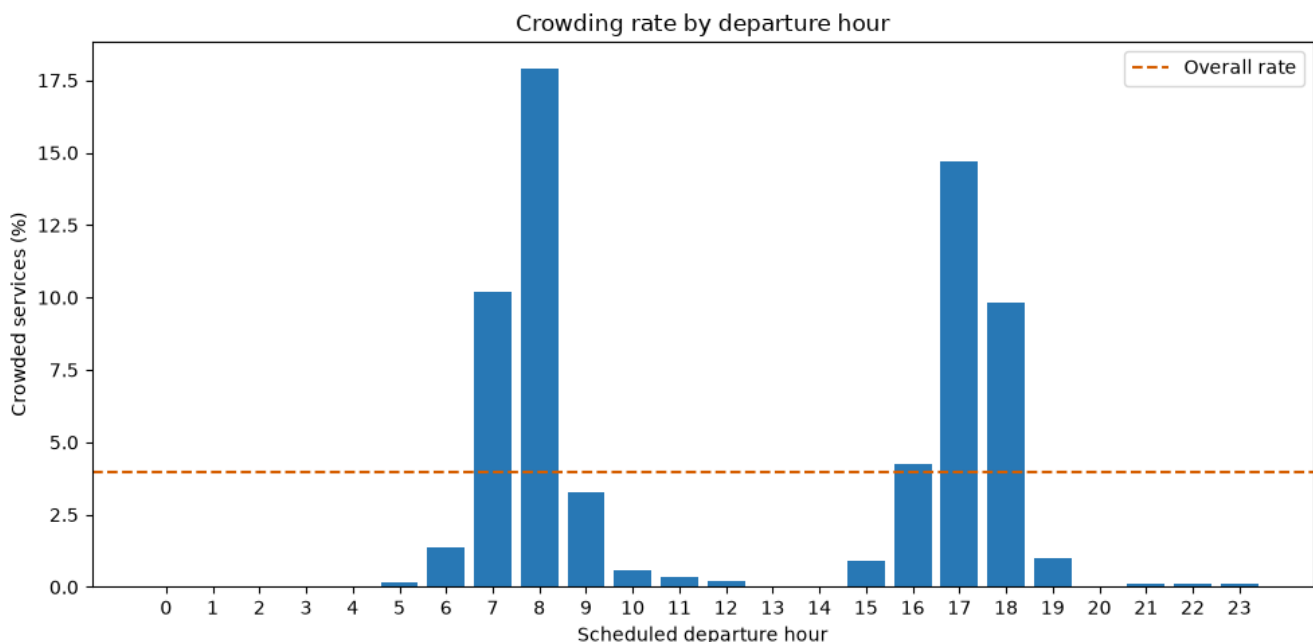
- Arrival and departure times had invalid or inconsistent format for example: 9.08 pm and 1421.
- Timestamps were cleaned by and repaired using the date from the paired timestamp.
- The final dataset contains 49,927 rows and 16 columns including the cleaned timestamps (Arrival_Clean and Departure_Clean)

4. Exploratory Data Analysis

There were 3 questions posed, and their corresponding figures and explanations are as follows:

I. Crowding Rate Per Hour:

At which departure hour is the crowding rate the highest?



Departure_Hour	Records	Crowding_Rate
8	3270	0.179205
17	3169	0.147050
7	2956	0.102165
18	3048	0.098425

Departure_Hour	Records	Crowding_Rate
16	2869	0.042524

We grouped the records by the departure hour and defined the crowded service as the ‘FEW_SEATS_AVAILABLE’ or ‘STANDING_ROOM_ONLY’. The chart displayed each hour’s crowded rate.

We found that the Crowding is highest during the commuter peak. I.e during start of day and end of day for average work shift. The identified departure hours are shown below with their corresponding crowding rate. The time range of morning(7am-8am) and afternoon(4pm-6pm) show the highest crowding.

II. Top 15 stations by crowding rate:

Which stops have the highest crowding risk?

To avoid unstable percentages from very small samples, we included only stations with at least 100 records. The top 15 stations are shown below:

Stop Station	Records	Crowding rate
Milsons Point	409	0.183374
North Sydney	420	0.142857
Wollstonecraft	384	0.140625
Redfern	1518	0.137022
Parramatta	390	0.128205
Waverton	353	0.124646
Tempe	168	0.119048

Green Square	301	0.116279
Ersleville	165	0.115152
Mascot	306	0.114379
Artarmon	387	0.113695
St Leonards	407	0.113022
Wolli Creek	761	0.103811
Sydenham	488	0.102459
Central	1973	0.099848

Milsons point has the highest crowding rate (18.3%), followed by North Sydney (14.3%), Wollstonecraft(14.06%), Redfern(13.7%) and Parramatta(12.8%)

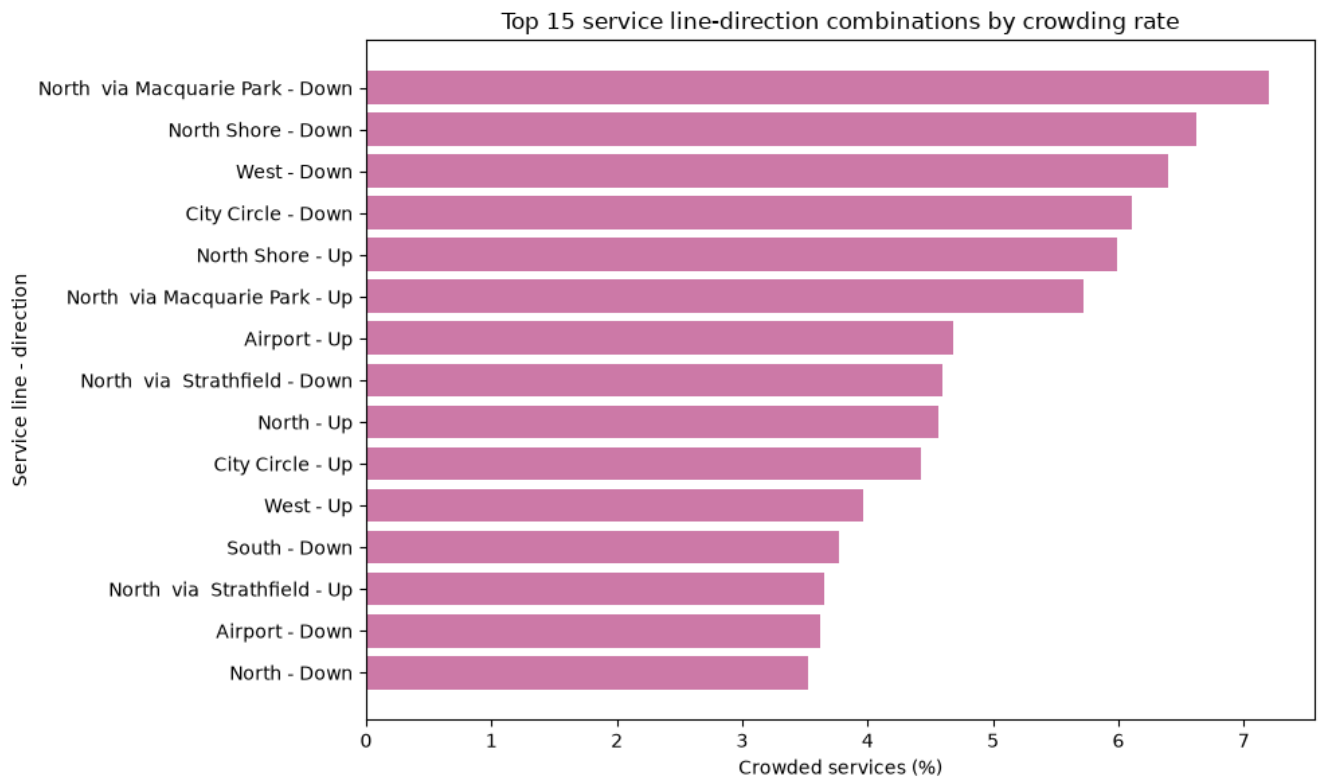
These station should be kept as the priority for investigating crowding issues. This analysis identified which station have the highest crowding rate, this does not mean that the stations themselves are the cause and hence a targeted investigation will be required to find the root cause. However this does show that these stations should be investigated first.

III. Top Service line-direction Combinations

Which line-direction combinations show the highest crowding rate?

We grouped records by service line and direction (Up or Down). We included only service line-direction combinations with at least 100 records, because smaller groups can produce unstable crowding percentages.

The findings are as follows:



Service.Line	Segment.Direction	Records	Crowding_Rate	Line_Direction
North	Down	1420	0.035211267605633804	North - Down
Airport	Down	2267	0.03617115130127922	Airport - Down
North via Strathfield	Up	793	0.03656998738965952	North via Strathfield - Up
South	Down	1804	0.037694013303769404	South - Down
West	Up	3002	0.039640239840106596	West - Up
City Circle	Up	1855	0.044204851752021566	City Circle - Up
North	Up	1336	0.04565868263473054	North - Up
North via Strathfield	Down	913	0.04600219058050383	North via Strathfield - Down
Airport	Up	2287	0.04678618277219064	Airport - Up

North via Macquarie Park	Up	1572	0.05725190839694656	North via Macquarie Park - Up
North Shore	Up	2304	0.059895833333333336	North Shore - Up
City Circle	Down	1311	0.06102212051868802	City Circle - Down
West	Down	3207	0.06392266916120985	West - Down
North Shore	Down	2477	0.0662091239402503	North Shore - Down
North via Macquarie Park	Down	1513	0.07204230006609386	North via Macquarie Park - Down

We found that the North via Macquarie Park–Down has the highest crowding rate (7.2%), followed by North Shore–Down (6.6%), West–Down (6.4%), and City Circle–Down (6.1%)

These lines should be the first candidates for the detailed service capacity reviews as they show the most crowding rates of all line-direction combinations. These are crucial for analysis since they represent the most crowding rates depending on the specific direction of the train lines. The NSW team can use this finding to further investigate the root causes and hence come up with targeted approaches such as checking if this is true throughout the day, weekends or is concentrated to a certain time.

Final Findings:

- Crowding is more prevalent in morning and afternoon peaks
- Crowding differs by station and service direction
- High-risk line-direction results might be due to small number of station time combinations

5. Assignment 2 recommendations

Assignment 2 could predict the 'Crowded' outcome using stop station, service line, direction, origin, destination, train-set type, and station-sequence order. Departure-time variables may be used only if they are available at the intended prediction time. 'Occupancy Range' should be excluded because it is

closely related to the outcome and could cause target leakage. The cleaned dataset provides the starting point for this later modelling work.

Limitations:

1. Data only covers a short period of time from 9-16 January
2. Findings are not covering other seasons, holidays, disruptions etc
3. Some categorical fields have missing values
4. EDA shows only the association of data and does not give a root cause
5. Crowded records are less with respect to the entire data set
6. Station and line combinations do not reflect peak time completely and hence a different combination might be required

GENAI disclosure

Generative AI (OpenAI Codex/ChatGPT) was used to support code explanation, reproducibility documentation, data-cleaning troubleshooting, and AI-assisted proofreading. All dataset checks, cleaning decisions, code outputs, visualisations, and interpretations were reviewed by the group against the raw data, cleaned data, unit materials, and assignment requirements. The group remains responsible for the final submitted work.

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